Lesson 5

Lenses

Objectives

The students will be able to:

- Understand how to draw ray diagrams for convex lenses.
- Understand how to solve problems involving convex lenses.
- Understand how to draw ray diagrams for concave lenses.

1. What are Lenses?

A lens is an optical device that is used to bend light in a specific way. It is a piece of clear plastic or glass with curved surfaces. Light is refracted as it enters and exits the lenses, and therefore changing the path of travel.

Lenses are classified into two types depending on how they bend rays of incident light. A converging lens bends light so that the light rays come together to a point while a diverging lens bends light so it spreads light apart instead of coming together.

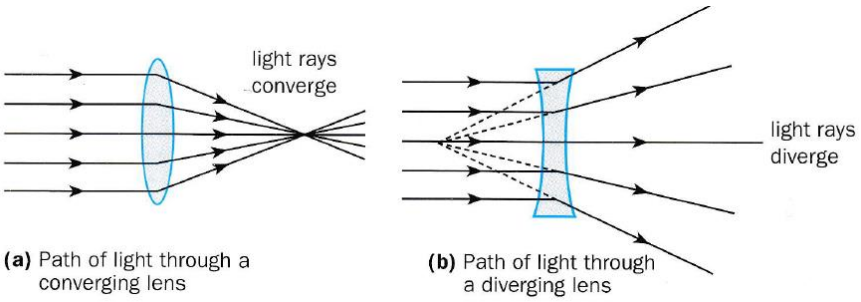


A peephole is an example of a diverging lens. It is a small security device that provide a panoramic view of objects and environments outside doors or walls. The view is created through the use of one or more concave lenses inside the device which minimizes the proportions of specific objects and gives a wide overview of an entire



The magnifying glass represents one of the simplest, most direct applications of a converging lens. As light enters the lens, it becomes focused to a specific focal point in front of the center of the lens.

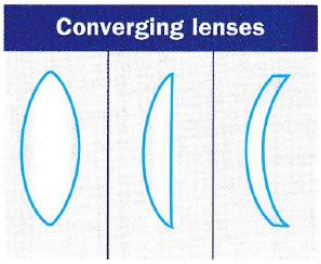
Any ray of light that is incident into lenses undergoes double refractions. Since both sides surfaces of lenses are not parallel to one another, light rays either converge or diverge after passing through the lenses.



(a) Path of light through a converging lens

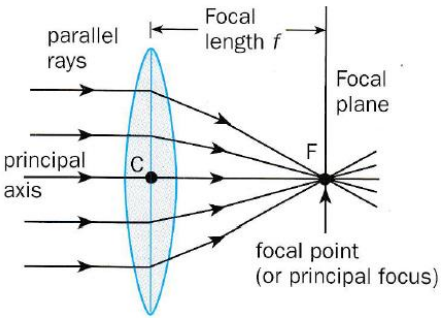
(b) Path of light through a diverging lens

2. Converging Lenses (convex lenses)

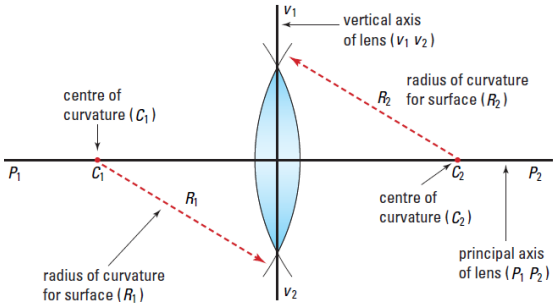


A **Converging Lens** is any lens that is thicker in the center than its edges. Converging lenses make rays, that parallel to the principal axis, converge at the focal point as shown below.

2.1 Properties of converging lenses



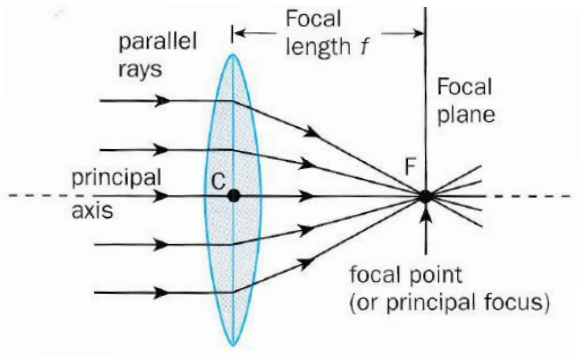
When you examine the simple diagram of a converging lens in Figure, you can see many similarities between mirrors and lenses. The term, convex, means curved outward for both mirrors and lenses, which is why converging lenses is sometime referred to as convex lenses.



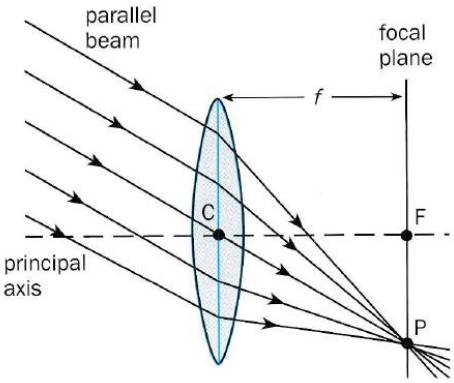
The surfaces of simple mirrors and lenses are spherical and described by radii of curvature. As well, any line passing through the geometric center of the lens is called an *axis*. The axis that passes through the center of curvature of the lens is the **principal axis**. The upright axis that is perpendicular to the principal axis is the **vertical axis**. All other axes are called **secondary axes**.

The point of convergence of the rays that enter parallel to the principal axis is called the **principal focus** or **focal point**. Since the rays of light can enter the lens from either side, there is a principal focus or focal point on the principal axis on each side of the lens. The distance from the vertical axis to either of the focal points is called the **focal length** of the lens. The intersection point between the principle axis and the vertical axis is the **optical center**. A plane that passes through the focal point and perpendicular to the principal axis is known as the **focal plane**. Parallel beam of rays (regardless if they are parallel to the principle axis or not) are refracted to a point on the focal plane.

(a). Beam of incident lights parallel to the principle axis.

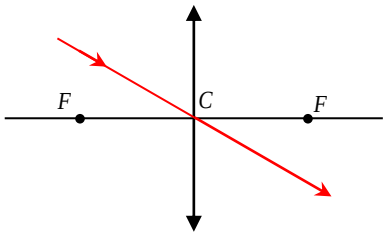
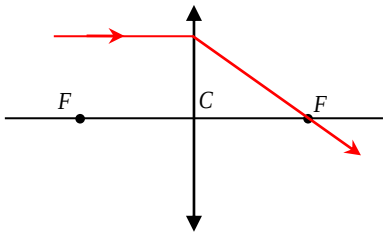
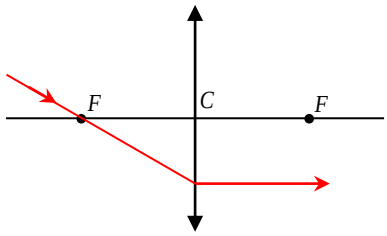


(b). Beam of incident lights at an angle to the principle axis.



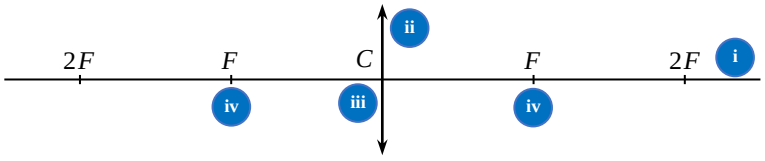
2.2 Ray diagrams for converging lenses

- Light ray1: any ray parallel to the principal axis will refract passing through the focal point.
- Light ray 2: any ray passing through the center of the lens will go straight.
- Light ray 3: any ray passing through the focal point will refract parallel to the principal axis.

Ray 1 passes through the optical center *C*	Ray 2 parallel to the principle axis	Ray 3 passes through the focal point *F*
		

How to draw ray diagram for thin converging lenses for objects placed further than the focal length

Step 1:



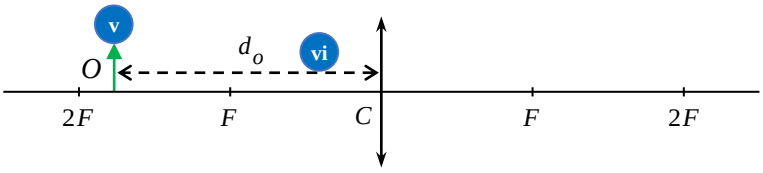
i. Draw a horizontal line to represent the principle axis.

ii. Draw a double-headed arrow perpendicular to the horizontal line to represent the converging lens.

iii. The intersection point between the principle axis in (i) and the lends in (ii) is the optical center of the converging lens. Label the point *C*.

iv. Label the focal point F of the lens on the principle axis. The distance *CF* is the focal length.

Step 2:

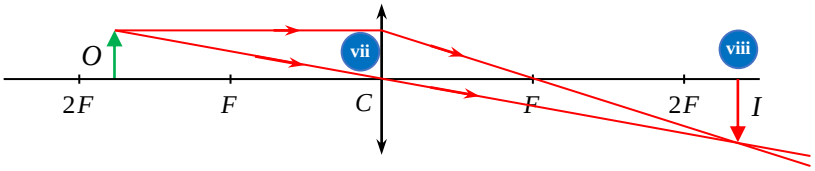


v. Draw a vertical arrow on the left of the lens to represent an object. Label the object *O*.

vi. Label the distance *OC* as d_o.

Step 3:

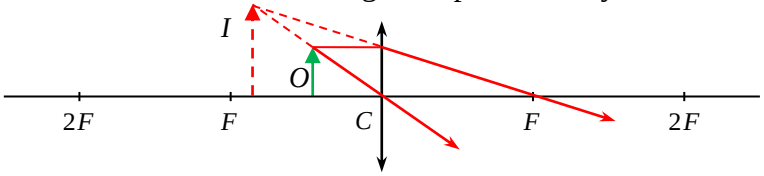
vii. Select and draw any two of the three rays from the tip of the object (in the diagram, we draw ray 1 and ray 2)



viii. The intersection point of the two rays represents the real image of the tip of the object. Complete the real image by drawing an arrow and labelling it *I*.

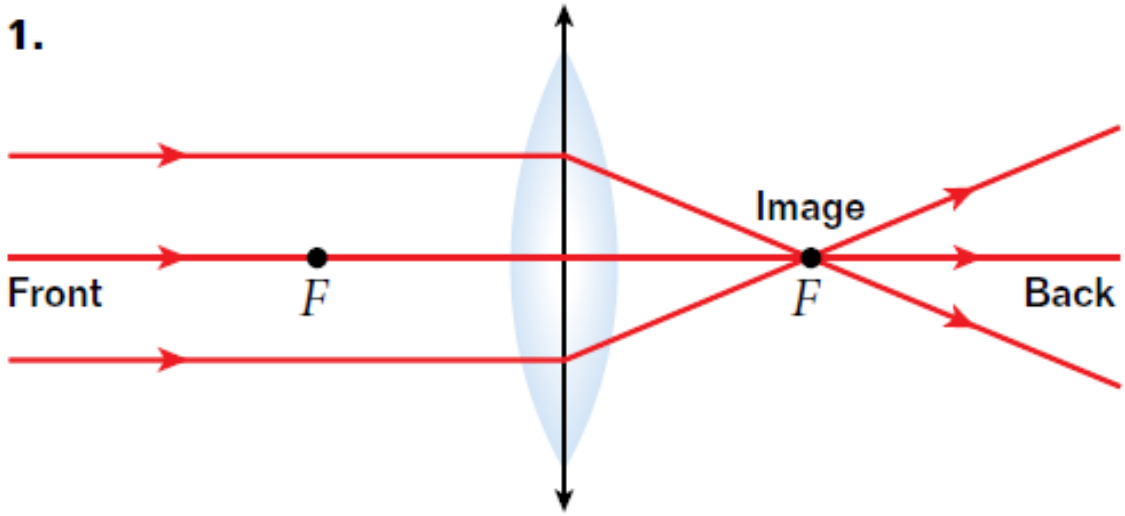
How to draw ray diagram for thin converging lenses for objects placed closer than the focal length

We can see that the light rays diverge and the intersection point of ray 1 and 2 can be found only if the rays are extended backwards. The virtual image is represented by a dashed arrow.



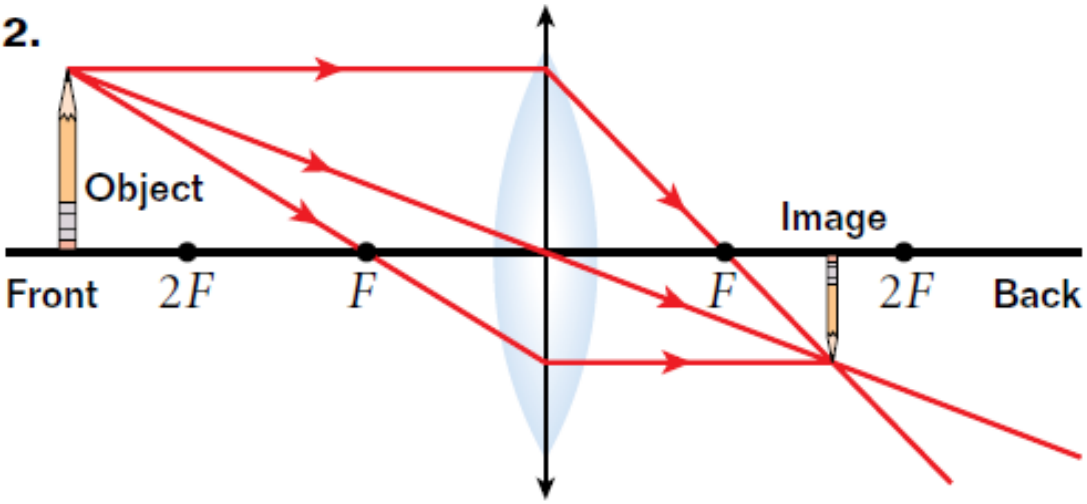
2.2 Images formed by converging lenses

Case 1: Object at infinity



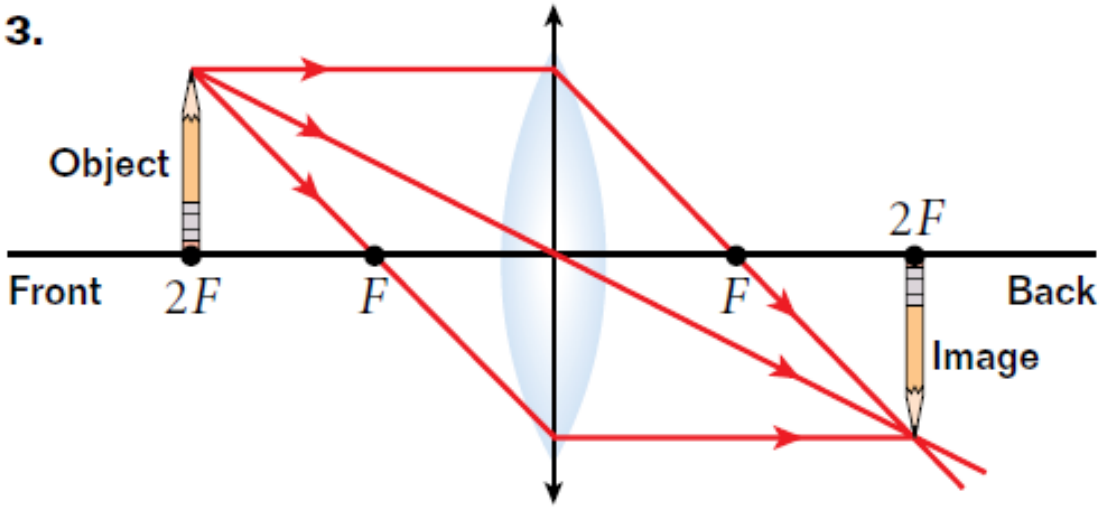
d_o	d_i	Image	Applications
$d_o = \infty$	$d_i = f$	- Inverted - Real - Diminished	Object lens of a telescope

Case 2: Object further than 2F



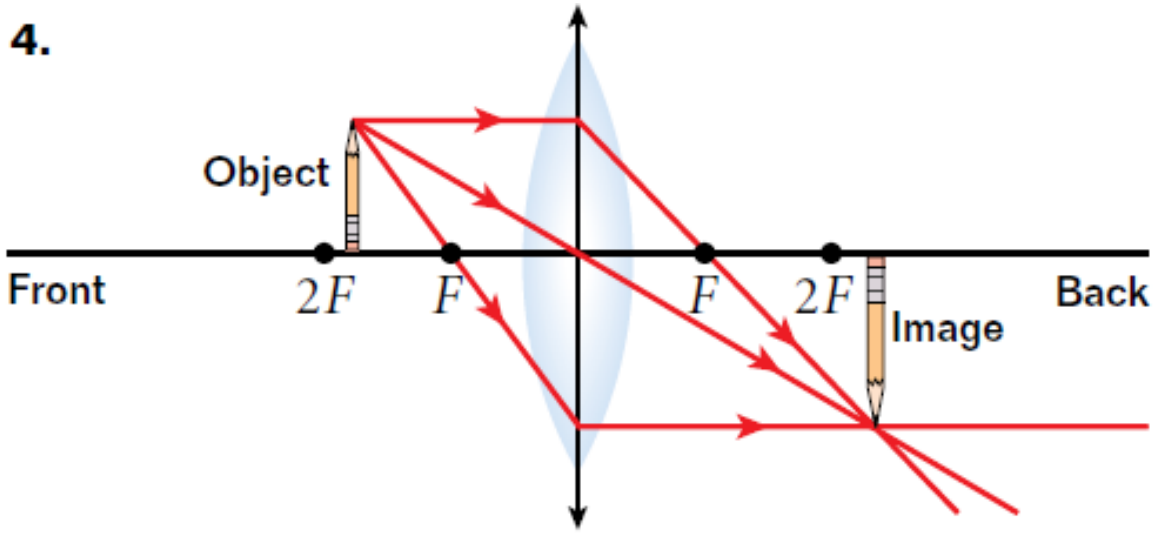
d_o	d_i	Image	Applications
$d_o > 2f$	$f < d_i < 2f$	- Inverted - Real - Diminished	- Camera - Eye

Case 3: Object at 2F



d_o	d_i	Image	Applications
$d_o = 2f$	$d_i = 2f$	- Inverted - Real - Same size	Photocopier making same-sized copy

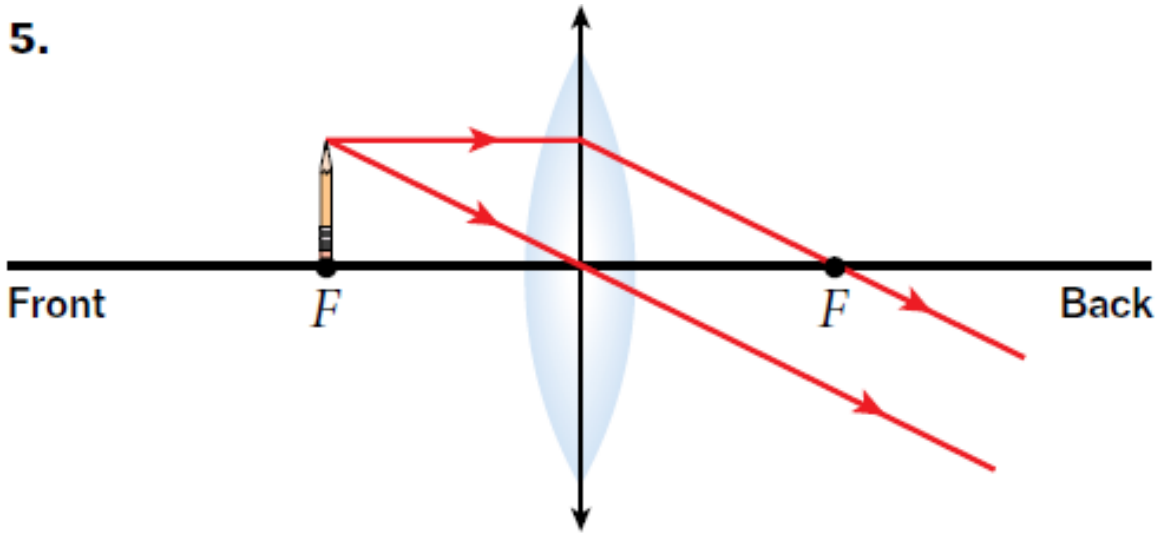
Case 4: Object 2F between F



d_o	d_i	Image	Applications
$f < d_o < 2f$	$d_i > 2f$	- Inverted - Real - Magnified	- Projector - Photograph enlarger

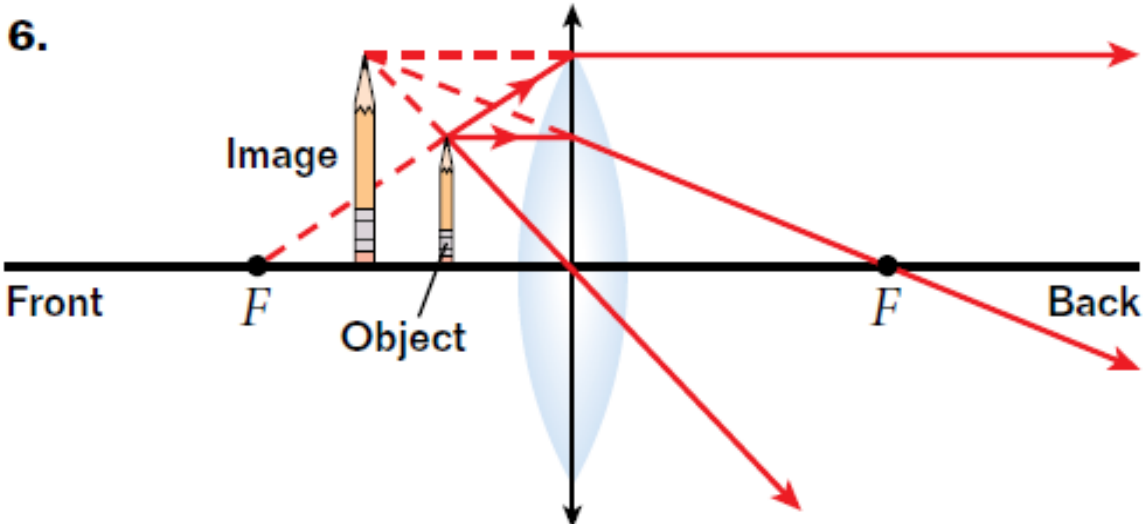
So far, the image formed by the converging lens is always a real image as we move the object from infinity to near the focal point. To refresh, a real image is an image formed by converging light beams. What happens when the object is moved even closer to the lens?

Case 5: Object at F



d_o	d_i	Image	Applications
$d_o = f$	$d_i = \infty$	- Upright - Virtual - Magnified	Eyepiece lens of a telescope

Case 6: Object between F and C



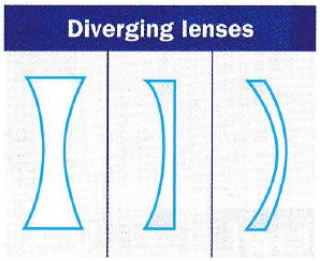
d_o	d_i	Image	Applications
$f < d_o$		- Upright - Virtual - Magnified	Magnifying glass

From the drawings above, we can see that

- when $d_o > f$, the image formed is real, inverted, and on the opposite side of the lens as the object.

- when $d_o \leq f$, the image formed is virtual, upright, and on the same side of the lens as the object.

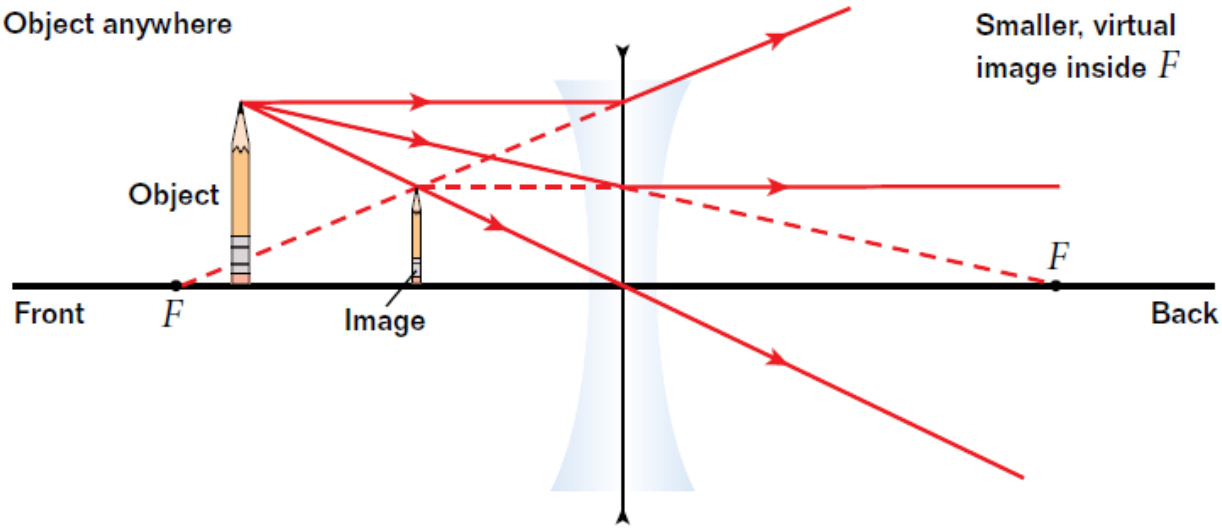
3. Diverging Lenses (concave lenses)



A **diverging lens** is any lens that is thinner at the center than its edges.
Diverging lenses make rays parallel to the principal axis diverge.
Diverging lenses can sometimes be referred to as concave lenses.

3.1 Images formed by converging lenses

The ray diagrams used to locate images are drawn much like those for converging lenses — with one major difference. Rays parallel to the principal axis of a converging lens are refracted to pass through the focal point on the opposite side of the lens. For a diverging lens, the rays parallel to the principal axis are refracted so that they appear to be diverging from the focal point on the side of the lens where they began.



4. Equations for Thin Lenses

Equations for lenses are the same as for curved mirrors, which shouldn't be surprising since lenses are essentially two curved mirrors glued together (though light refracted rather than reflected).
The relationship between the object's location and the image's is given by

$$\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i}$$

The magnification of a thin lens is

$$M=\frac{h_i}{h_o}=-\frac{d_i}{d_0}$$

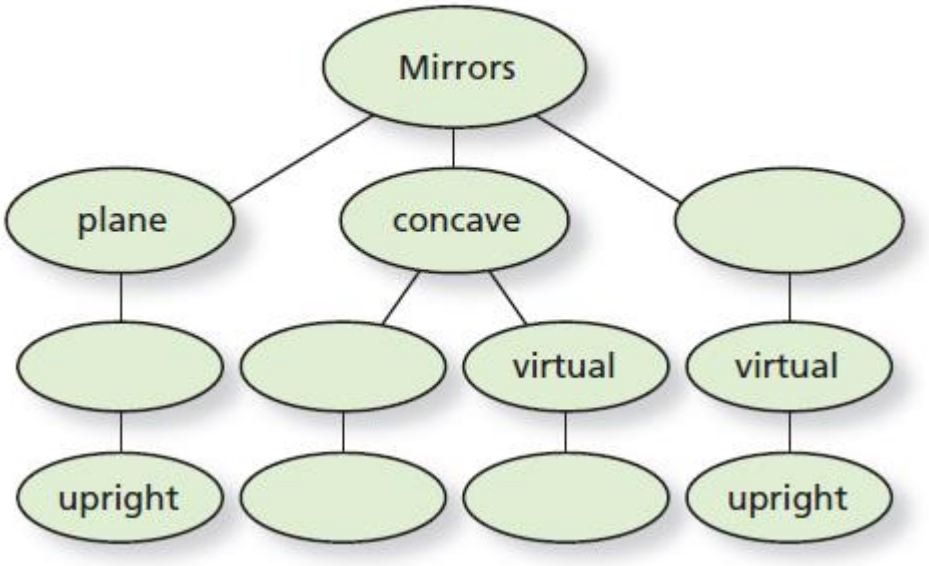
Sign conventions for lenses

As with curved mirrors, some quantities of lenses are sign sensitive. The value of the focal length, for instance, is positive for converging lenses and instead negative for diverging lenses.

	Positive value	Negative value
d_o	Object is in front of the lens	Object is behind the lens
d_i	Image is behind the lens	Image is in front of the lens
h_o	Upright	Inverted
h_i		
f	Converging lens	Diverging lens

Practices

1. Complete the following concept map using the following terms: *convex, upright, inverted, real, virtual*.



2. A _____ is an optical instrument that forms images by refraction.

3. Lenses that are thicker in the middle are called _____ lenses (Fig 3a).

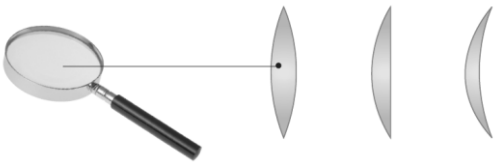


Fig 3a

4. Lenses that are thinner in the middle are called _____ lenses (Fig 3b).

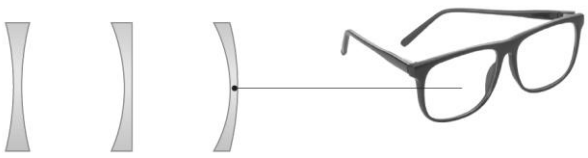


Fig 3b

5. A _____ lens bends the light inwards, or converges the light (Fig 3c). Therefore, it is called a _____ lens.

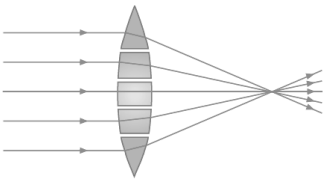


Fig 3c

6. A _____ lens bends the light outwards, or diverges the light (Fig 3d). Therefore, it is called a _____ lens.

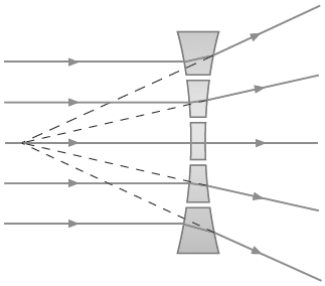
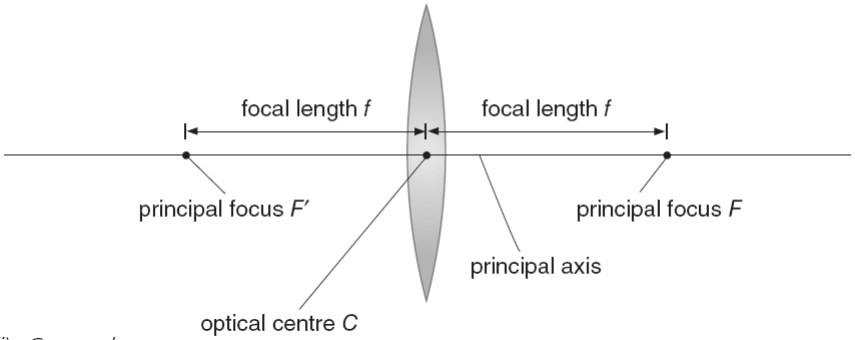


Fig 3d

7. A lens (convex or concave) has some key features (Fig 3e).



(i) Convex lens.

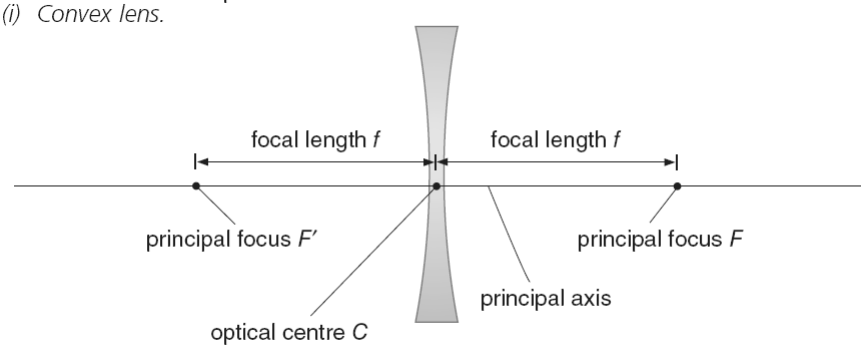
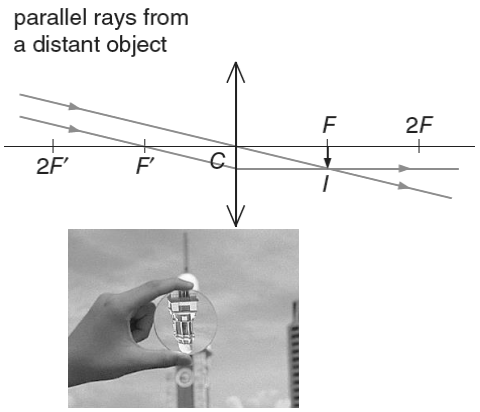
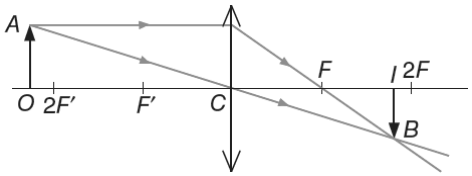
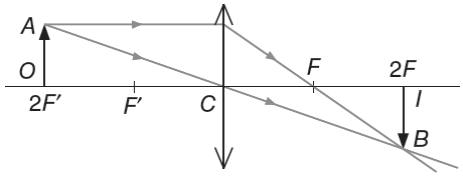
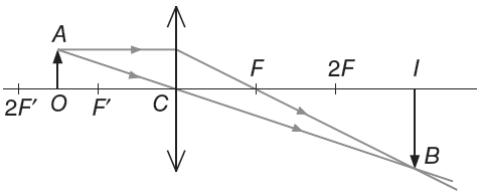


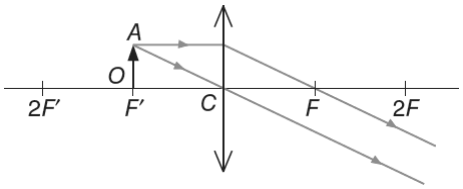
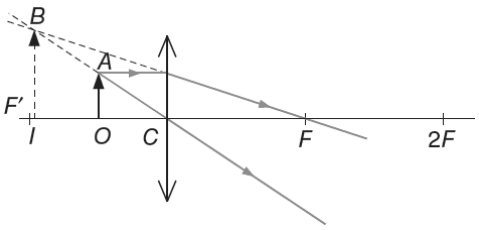
Fig 3e (ii) Concave lens.

Optical centre (C)	The centre of the lens. Light rays passing through it _____ change direction.
Principal _____	The line passing through the optical centre and perpendicular to the lens.

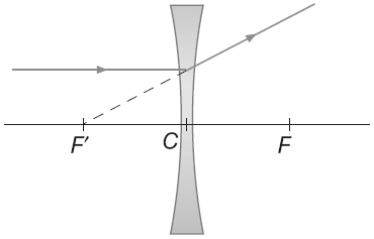
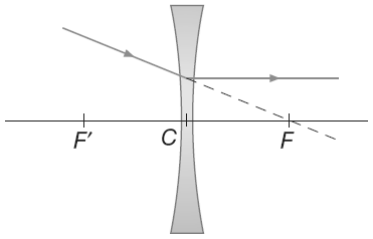
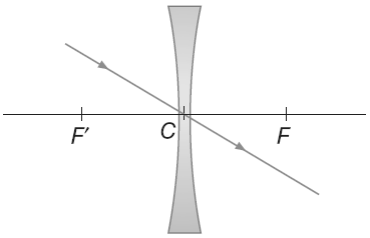
Principal _____ (F or F')	The point that rays parallel to the principal axis converge to (for a convex lens) or appear to diverge from (for a concave lens) after passing through the lens. Each lens has two principal foci, one on each side of it.
_____ (f)	The distance of the focus from the optical centre. A thicker convex lens converges light more and therefore has a _____ focal length. Similarly, a concave lens with a larger curvature diverges light more, so it also has a _____ focal length.

8. A ray diagram can help us locate an image and find out its nature.

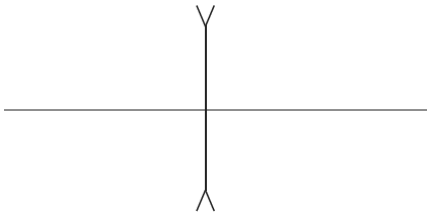
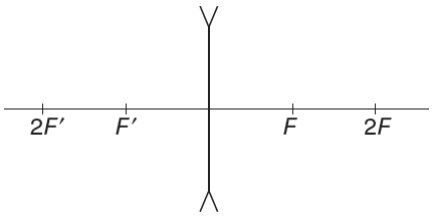
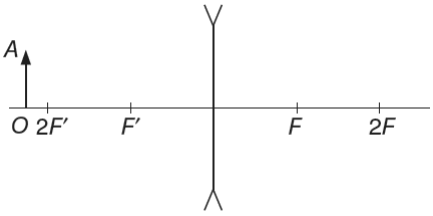
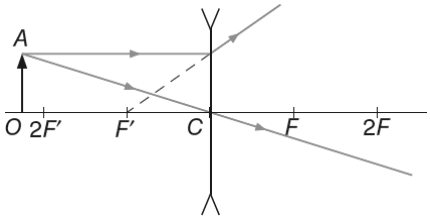
Object position	Ray diagram	Image properties
Distance object (object at infinity)		Position: _____ _____ Nature: _____ _____ _____
Object beyond 2*F'*		Position: _____ _____ Nature: _____ _____ _____
Object at *2F'*		Position: _____ _____ Nature: _____ _____ _____
Object between *F'* and 2*F'*		Position: _____ _____ Nature: _____ _____ _____

Object at *F'*		Refracted rays are parallel. Position: _____ _____ (i.e. no image formed)
Object within *F'*		Position: on the same side of the object Nature: _____ _____ _____

9. The construction rules for a concave lens can be summarized as follows:

Rule 1 A ray parallel to the principal axis is bent so that it appears to come from the _____ (*F'*).	
Rule 2 Rays appear to pass through the focal point will refract parallel to the principal axis after refraction.	
Rule 3 A ray passing through the _____ travels straight on.	

10. The steps for drawing a ray diagram for an object placed behind 2*F'* of a concave lens are as follows:

Step 1 Draw a horizontal line to represent the _____ axis and draw the symbol for a concave lens.	
Step 2 Mark the positions of *F'*, 2*F'*, *F* and 2*F* on the principal axis.	
Step 3 Draw the object as an _____ standing upright on the principal axis at the given position.	
Step 4 Draw rays, with arrows to indicate directions, from the tip of the object using any two of the construction rules. Use dotted lines for extending rays _____ to locate the virtual image behind the lens.	
Step 5 Draw a dotted arrow as the virtual image with its tail on the _____	